

Nihed Harbi

Control & Automation Engineer

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Control & automation engineer trained in modeling, control, industrial automation, and robotics, with hands-on experience in R&D technical support within the home appliance industry (product validation, regulatory compliance, continuous improvement). Rigorous, autonomous, and problem-solving oriented, I combine discipline and team spirit to deliver demanding technical projects.

Professional Experience

R&D Technical Support Engineer – BIOLUX (Beko & Maxwell)

September 2025 – Present

- Validated and functionally tested 15+ home appliance references in SKD production, achieving a 100% compliance rate with EN 60335, EN 60456, and EN 50242 standards
- Oversaw the entire industrialization process of CKD refrigerators, including quality/compliance control and fault diagnosis at each manufacturing stage
- Applied LVD, EMC, RoHS, and Ecodesign regulatory requirements across all product validation activities
- Diagnosed and resolved dozens of non-conformities using 8D, 5 Whys, and Ishikawa (Fishbone) methodologies, implementing corrective and preventive actions (CAPA)
- Contributed to the industrialization of 3+ new product references by applying Lean Manufacturing, 5S, and PDCA principles, achieving an 18% reduction in scrap rates

Intern – SONEGAL Algerian Power Production Company SPE

March 2023 – April 2023

- Studied DCS systems, PID control loops, and industrial control architectures
- Operated HMI interfaces for process supervision
- Supported maintenance of instrumentation equipment

Projects

Master's Thesis: Kinematic and dynamic modeling, control, and motion planning of a 7-DOF Franka Emika Panda robotic manipulator under MATLAB/ROS for a pick-and-place application (real-time execution)

Project: Design and implementation of a three-threshold temperature control system for an egg incubator with ESP32-based supervision (± 0.5 °C)

Bachelor's Thesis: Design and implementation of an Arduino-based automated regulation system for an agricultural greenhouse, with adaptive day/night threshold logic and actuators control

Internship Project: Programming of a water filtration regulation system under TIA Portal, simulated in Factory I/O and supervised by a SCADA system

Project: Position control of a servo motor under STM32 via PID and state feedback(20 ms, $\pm 2^\circ$)

Skills

Control Systems: Modeling, identification, optimization, real-time control development

Industrial Automation: Control system architecture design, SCADA development, HMI/API interface development

Robotics: Robot kinematic and dynamic modeling, motion planning, perception, navigation

Embedded Systems & Electronics: PCB design, FPGA, microcontroller programming (STM32, ESP32, Arduino)

Programming: VHDL, Assembly, C/C++, Python, VBA, PLC programming (IEC 61131-3)

Communication Protocols: Modbus, PROFINET, CAN Bus, HART, TCP/IP, MQTT

Software Tools: MATLAB/Simulink, NI LabVIEW, TIA Portal, Factory I/O, ISIS Proteus, SolidWorks, AutoCAD, ROS2

Education

Master's Degree in Automatic Control | Major: Automatic & Industrial Computing

2023 – 2025

University of Mentouri Brothers Constantine 1

Bachelor's Degree in Automatic Control

2021 – 2023

University of Mentouri Brothers Constantine 1

Preparatory Classes in Engineering

2018 – 2021

National Polytechnic School of Constantine

Certifications

MathWorks Certifications (MATLAB): Deep Learning, Machine Learning, Wireless Communications, MATLAB Coder, MATLAB Onramp

Languages

French – B2 TCF • English – C1 Advanced EF SET • Arabic – Native